

Imane El Abid

DISTRIBUTED SYSTEMS & BLOCKCHAIN RESEARCHER

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Summary

Ph.D. in Computer Science specializing in distributed systems and blockchain technologies. My research focuses on designing scalable, fault-tolerant, and resource-efficient distributed protocols to address the challenges of decentralized computing in heterogeneous environments.

Skills

Programming: Python, Java, C, Solidity, Scala.

Distributed Computing: Blockchain, Consensus Protocols, Byzantine Fault Tolerance.

Networking & Security: TCP/IP, SDN, IoT, Network Virtualization, Cryptography, PKI, ZKPs.

Tools & Platforms: Docker, Git, MATLAB, PyTorch, LaTeX.

Languages: Arabic, English, French, Japanese.

Education

Ph.D. in Computer Science

BenGuerir, Morocco

COLLEGE OF COMPUTING, UNIVERSITY MOHAMMED VI POLYTECHNIC, (UM6P)

2020 - 2026

Thesis: *Towards Blockchain Systems for Resource-Constrained Devices: Scalability and Data Redaction*

Predoctoral Studies, DNA Program

BenGuerir, Morocco

UNIVERSITY MOHAMMED VI POLYTECHNIC, (UM6P)

2018 - 2019

Intensive predoctoral program focusing on Data, Networks, and Algorithms (DNA).

Network & Telecommunication Engineer

Marrakech, Morocco

NATIONAL SCHOOL OF APPLIED SCIENCES, (ENSA)

2012 - 2017

Specialized in: Network Design & Security, Network Virtualization and Distributed Systems Fundamentals.

Work Experience

Doctoral Researcher (PhD)

Benguerir, Morocco

COLLEGE OF COMPUTING (UM6P)

2020 - 2026

- Led doctoral research addressing core distributed systems problems, building solutions for fault tolerance and scalability within privacy-aware blockchain architectures.
- Designed and implemented POCKETCHAIN, a lightweight blockchain architecture integrating *PocketChain Consensus Protocol (Pccp)* with role specialization and an incentive model to improve scalability and fairness for secure asset transfers within *mobile and edge environments*.
- Designed and developed an extensible benchmarking framework for evaluating *Redactable Blockchain* techniques.
- Designed LIGHTx, a lightweight transaction dissemination framework combining *Byzantine Reliable Broadcast* with a *Proof-of-Bandwidth* and a reputation mechanism for secure and scalable agreement.
- Published findings across several peer-reviewed articles in Q1 journals and highly-ranked international conferences.

Research Intern

Casablanca, Morocco

MOBILINFO

Jan. 2017 - Jul. 2017

- Studied state-of-the-art solutions for consistent updates in *Software-Defined Networking (SDN)* architectures.
- Conducted in-depth protocol analysis across various network infrastructures, including *IoT*, *WSN*, and *NFC*.

TECH: SDN, Mininet, OpenFlow, Centralized Controllers.

Early Systems & Network Internships

Marrakech, Morocco

DATA EMBASSY | MENARA MARRAKECH AIRPORT

Summer 2015 - 2016

- Configured high-availability Asterisk IPBX telephony servers and developed failover backup solutions.
- Administered Windows Server 2012 environments and Exchange mail services, utilizing Active Directory for access control.

TECH: Asterisk, IPBX, SIP, H.323 | Windows Server 2012, Active Directory, Exchange.

Projects

Paxos-based blockchain prototype

DESIGN AND IMPLEMENTATION OF A PERMISSIONED BLOCKCHAIN PROTOTYPE USING THE PAXOS CONSENSUS ALGORITHM.

2019

- **TECH:** Blockchain, Paxos Algorithm, Python

IBE-based Data Deletion Protocol for Cloud

DESIGN AND IMPLEMENTATION OF AN IBE PROTOCOL ENSURING SECURE DATA DELETION IN CLOUD ENVIRONMENTS.

2018

- **TECH:** Java, Identity-Based Encryption (IBE), Cloud APIs, Security Protocols.

Secure P2P Chat Application

DEVELOPED A SECURE P2P CHAT APP WITH PYTHON, IMPLEMENTING END-TO-END ENCRYPTION OVER DIRECT SOCKET CONNECTIONS.

2018

- **TECH:** Python, Sockets, Cryptography

Distributed KNN

DESIGNED AND IMPLEMENTED A DISTRIBUTED K-NEAREST NEIGHBOR (KNN) ALGORITHM.

2018

- **TECH:** Java, Distributed Computing

Consistent Updates in Software-Defined Networks (SDN)

STUDY OF STATE-OF-THE-ART SOLUTIONS FOR CONSISTENT UPDATES IN SDN ARCHITECTURES FOR DYNAMIC AND FAULT-TOLERANT NETWORK UPDATES.

2017

- **TECH:** OpenFlow, Python, SDN Controllers.

Study of QoS-aware routing algorithms in Cognitive Radio Networks.

SIMULATION OF ALGORITHMS FOR OPTIMAL RESOURCE ALLOCATION IN COGNITIVE RADIO NETWORKS.

2017

- **TECH:** Cognitive Radio Networks, MATLAB, Optimization Algorithms.

5G SC-FDMA Simulation

STUDY AND SIMULATION OF SC-FDMA UPLINKS IN LTE NETWORKS UNDER VARIOUS CONDITIONS.

2016

- **TECH:** MATLAB, LTE Protocols, Signal Processing.

Conferences & Workshops

2024	IEEE Dapps , IEEE International Conference on Decentralized Applications and Infrastructures	Shanghai, China
2024	UM6P Science Week , Mohammed VI Polytechnic University	BenGuerir, Morocco
2023	SRDS , International Symposium on Reliable Distributed Systems	Marrakesh, Morocco
2023	IWCMC , IEEE International Wireless Communications and Mobile Computing	Marrakesh, Morocco
2022	AMLD , Applied Machine Learning Days Africa	BenGuerir, Morocco
2022	JDD'22 , UM6P Doctoral Day	BenGuerir, Morocco
2021	NETYS , International Conference on Networked Systems	Marrakesh, Morocco
2021	ICBC , IEEE International Conference on Blockchain and Cryptocurrency	Sydney, Australia
2021	JDD'21 , UM6P Doctoral Day	BenGuerir, Morocco
2019	METIS , International Spring School on Distributed Systems	Marrakesh, Morocco
2019	NASMA , North African Machine Learning Summer School	BenGuerir, Morocco

Publications

Beyond Immutability: A Comprehensive Review of Redactable Blockchain Systems

Computer Networks

EL ABID, I., BOUBOUH, K., BENKAOUZ, Y.

2026

PocketChain: Redefining Blockchain Integration With Resource-Constrained Devices

FGCS

EL ABID, I., BOUBOUH, K., BENKAOUZ, Y.

2025

A Comparative Analysis of Blockchain Redaction Techniques

IEEE DAPPS

EL ABID, I., BENKAOUZ, Y.

2024

LighTx: a lightweight transactions transfer system

ICBC

EL ABID, I., BENKAOUZ, Y.

2021

LighTx: A Lightweight Proof-of-Bandwidth Transactions Transfer System

NETYS

EL ABID, I., BENKAOUZ, Y., KHOUMSI, A.

2021